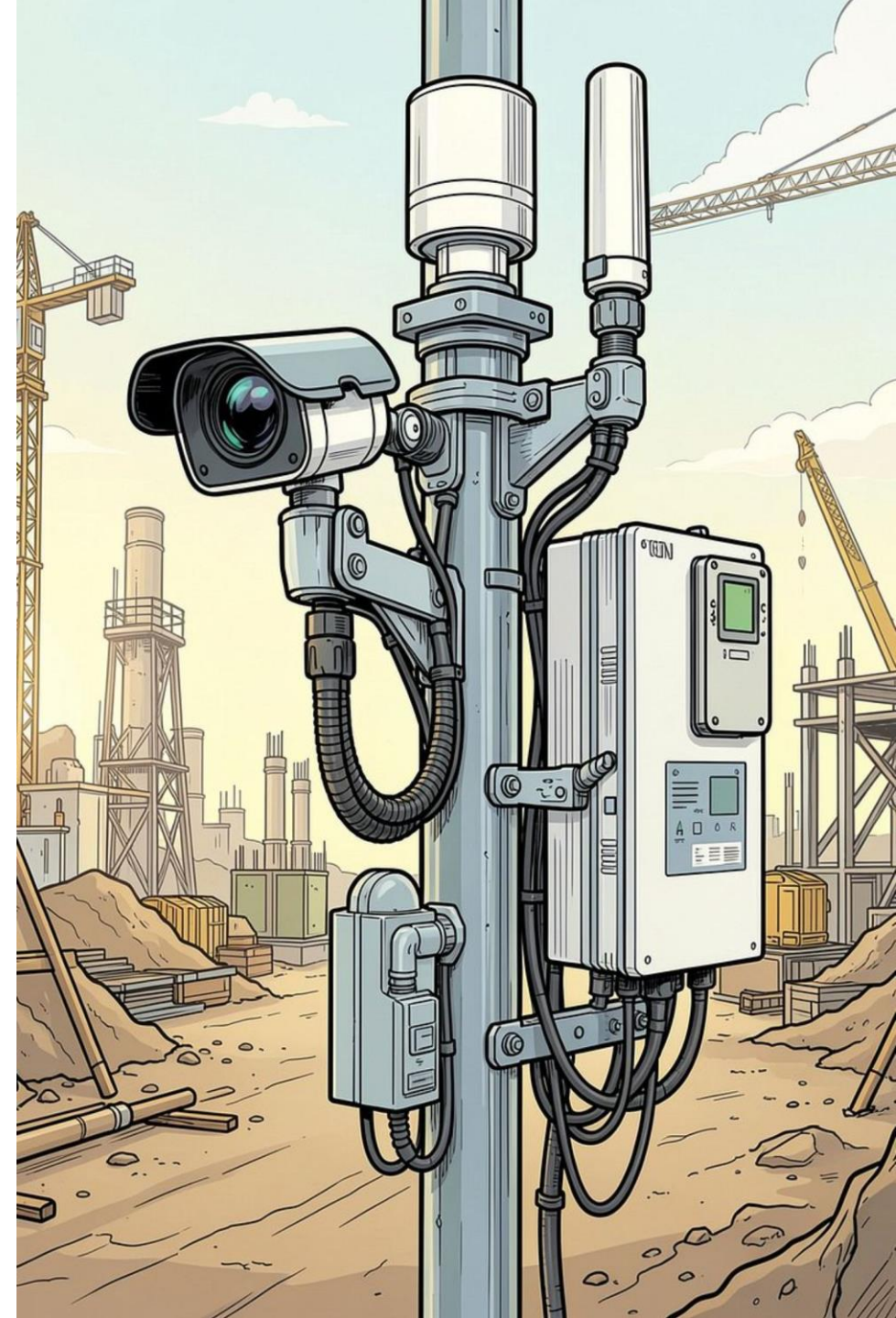


Software Engineering Laboratory

AI-Enabled IoT-Based Real-Time Environmental and Surrounding Safety Monitoring System SmartSurround

Intelligent Monitoring · Detection · Prediction · Safety



The Problem: Reactive Monitoring Isn't Enough

Traditional systems monitor one or two parameters, require manual observation, and only warn *after* unsafe conditions occur — leaving workers and the public exposed.

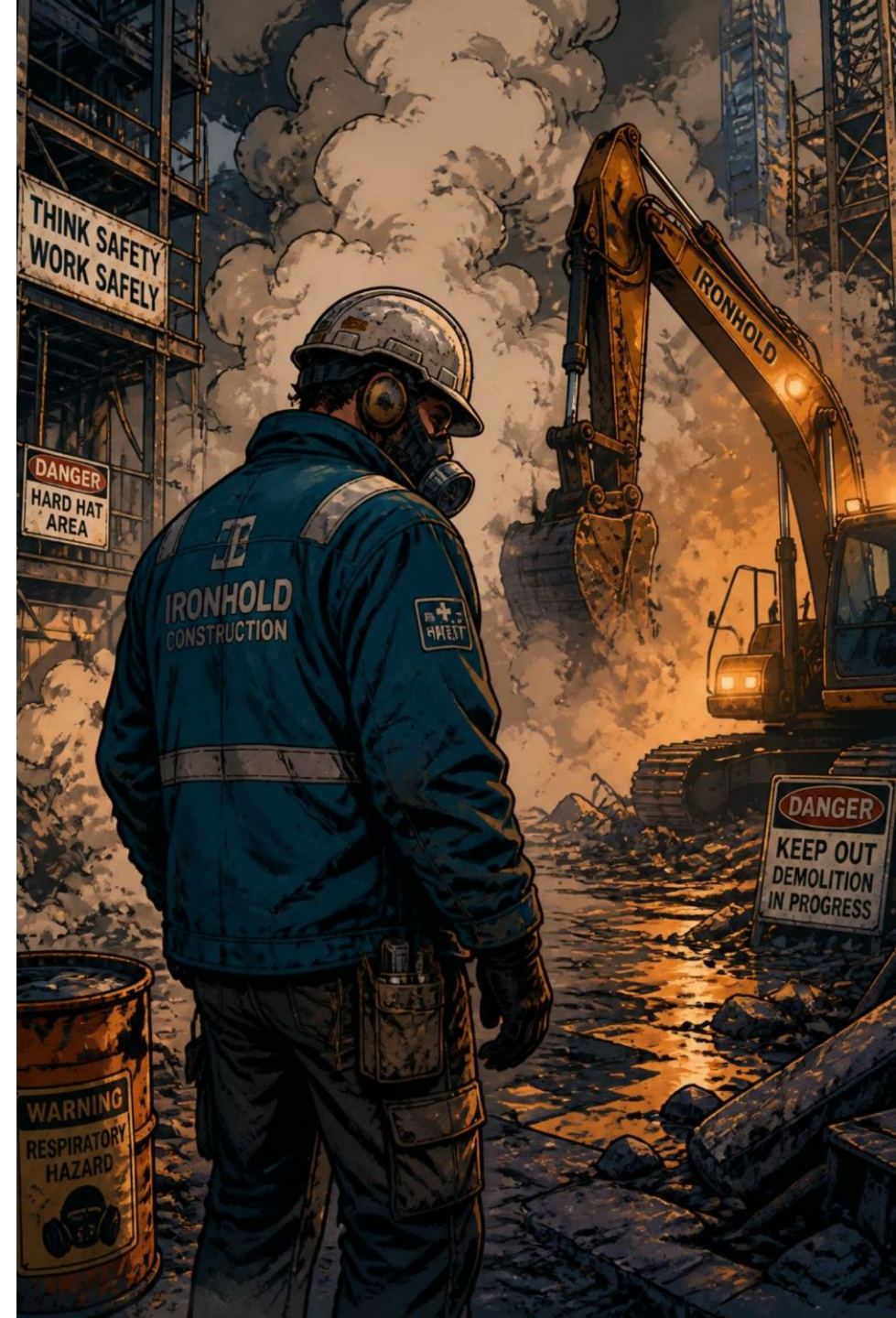
✗ Traditional Approach

Manual checks → Delayed detection → Risk exposure → Reactive response

✓ SmartSurround

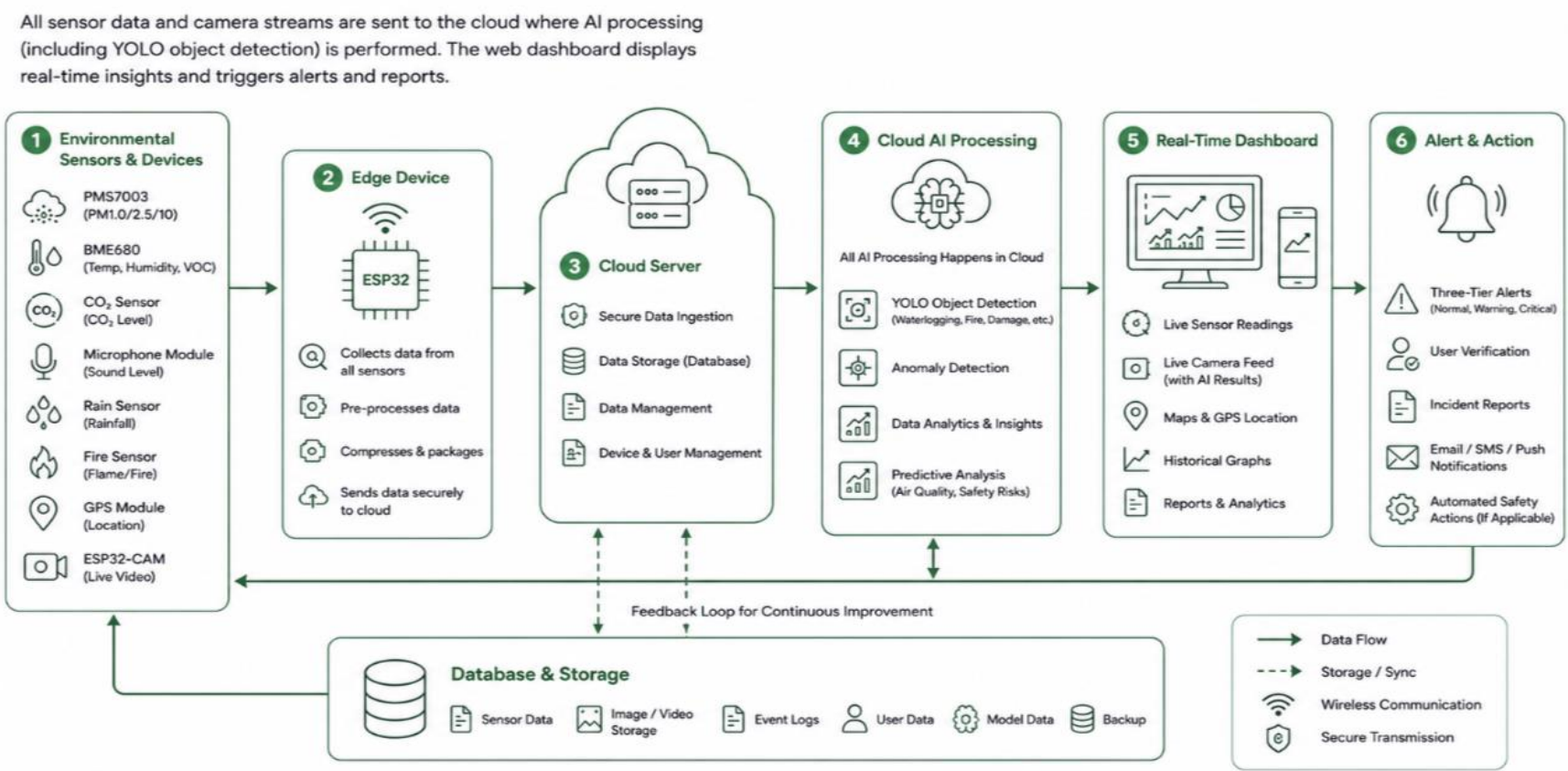
Continuous sensing → AI detection → Prediction → Proactive safety action

📄 **Core motivation:** The system should **understand** the environment, identify risks, predict problems, and help people act — not just collect data.



System Architecture & Working Flow

Sensors collect data, the ESP32 transmits it wirelessly, and a web dashboard displays real-time readings with AI analysis triggering alerts and incident reports.



The ESP32-CAM also enables AI-powered visual anomaly detection — identifying fire, waterlogging, damaged roads, and other hazards from live camera imagery.

Hardware Components & Sensing Parameters

IoT MONITORING POST

SMART ENVIRONMENTAL MONITORING SYSTEM

A multi-sensor IoT device for real-time environmental data collection, remote monitoring, and smart alerts.

1

GPS MODULE
Provides location tracking and geotagging for accurate monitoring.



2

PMS7003 PARTICLE SENSOR
Measures PM1.0, PM2.5, and PM10 concentrations.



3

BME680 ENVIRONMENT SENSOR
Monitors temperature, humidity, pressure, VOC, and air quality.



4

ESP32-CAM CAMERA
Live video streaming and AI-based visual analysis.



5

GPS ANTENNA
Enhances GPS signal reception for better accuracy.



6

RAIN SENSOR
Detects rainfall intensity and rain occurrence.



7

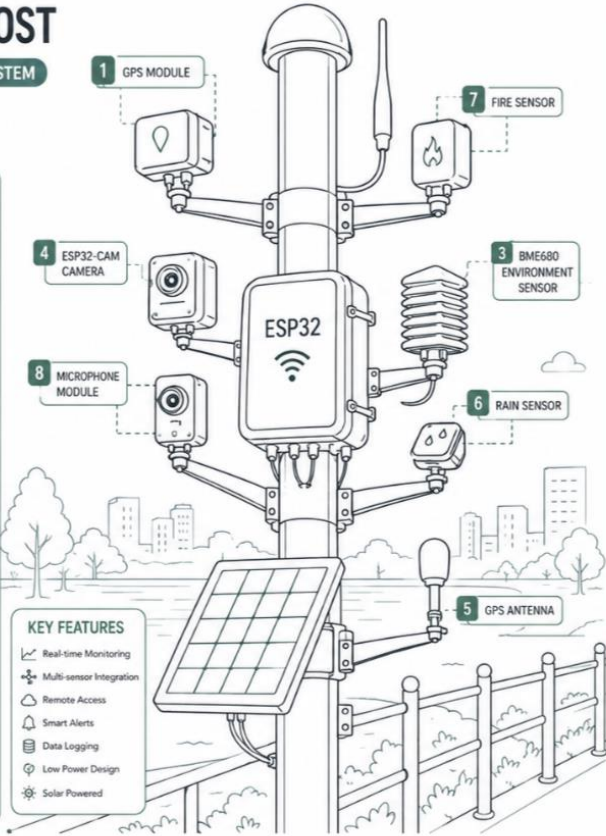
FIRE SENSOR
Detects fire and high temperature for early warning.



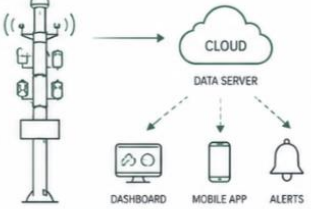
8

MICROPHONE MODULE
Measures ambient sound levels and noise monitoring.





SYSTEM OVERVIEW



DATA MONITORED

 PM1.0 / PM2.5 / PM10	 TEMPERATURE	 HUMIDITY
 CO ₂ LEVEL	 VOC LEVEL	 PRESSURE
 SOUND LEVEL	 RAIN STATUS	 FIRE STATUS

 GPS LOCATION

TECHNICAL SPECIFICATIONS

Microcontroller	ESP32-WROOM-32
Connectivity	Wi-Fi / Bluetooth
Power Supply	Solar Panel / Battery (3.7V)
Operating Temperature	-20°C to 60°C
Monitoring Parameters	12+
Data Storage	Cloud / SD Card (Optional)


IP65
WATER RESISTANT

ESP32	Main IoT controller & wireless comms
ESP32-CAM	Live video & AI visual detection
PMS7003	PM1.0, PM2.5, PM10
BME680	Temperature, humidity, VOC
CO ₂ Sensor	Carbon dioxide levels
Mic Module	Sound-level monitoring
Rain Sensor	Rain detection
Fire Sensor	Flame detection
GPS Module	Lat, long & location tracking

All components are mounted on a single monitoring post, continuously observing the surrounding environment.

AI-Powered Monitoring & Intelligent Web Dashboard



1 Real-Time Dashboard

Live readings for all sensors, GPS coordinates, and camera feed. Historical graphs track trends and flag anomalies.



2 AI Camera Analysis

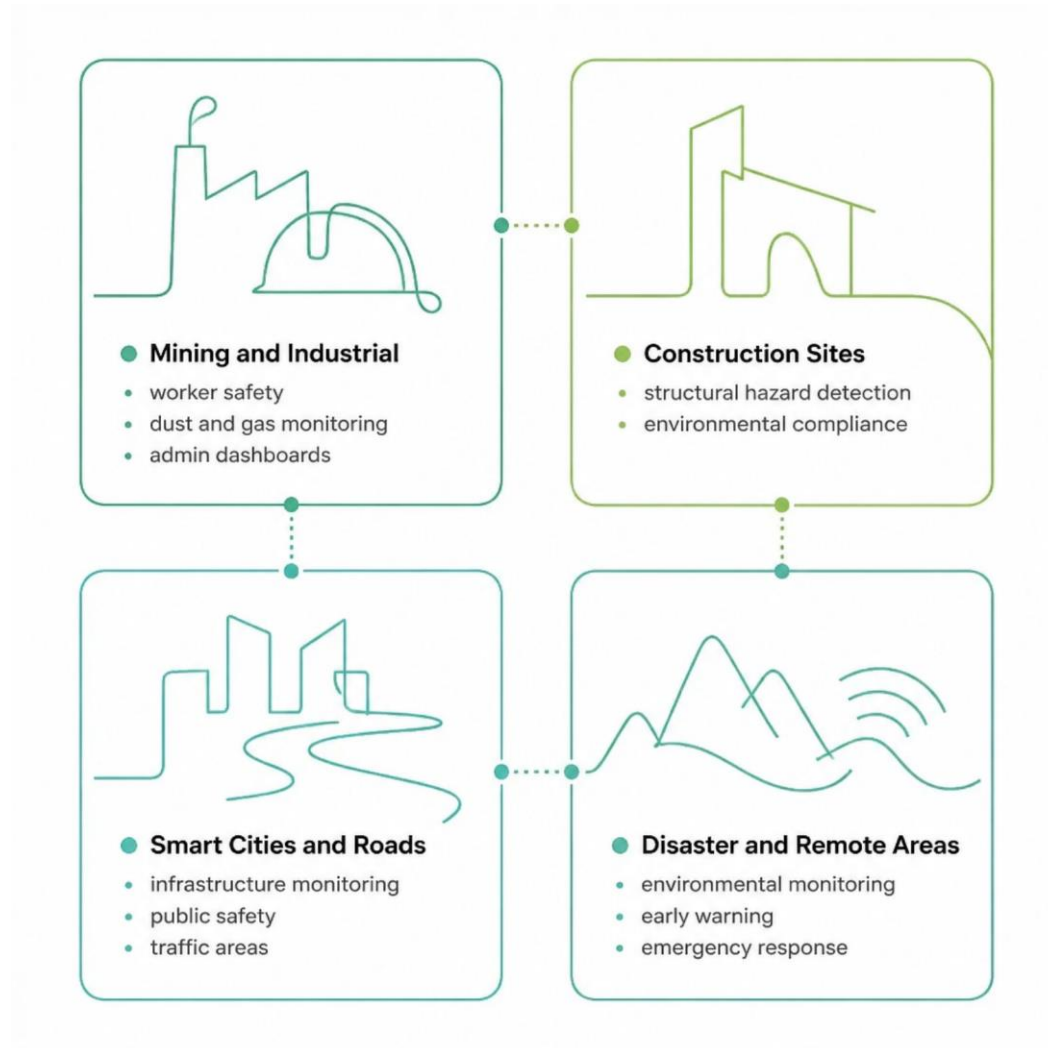
ESP32-CAM detects waterlogging, damaged roads, and fire/smoke. Detections require **user verification** before generating reports — reducing false alerts.



3 Intelligent Alert System

Three-tier alerts: **Normal** → **Warning** → **Critical**. Triggers dashboard warnings, notifications, and incident reports when thresholds are crossed.

Applications & Intelligent Safety Management



Role-Based Access

- **Admin:** Full dashboard access, threshold configuration, AI monitoring, reports
- **Worker:** Real-time conditions and safety warnings — no threshold modification

Predictive Safety Response

Historical data trains AI/ML models to predict risk before it escalates.

- **Example:** Rising dust + temperature → early warning → automated sprinkler/dust suppression → continued monitoring. Automated control uses only safety-rated hardware and predefined rules.

Expected Outcomes, Future Scope & Conclusion

Expected Outcomes

- Continuous real-time environmental monitoring
- AI-assisted visual anomaly detection
- GPS-based incident reporting with evidence
- Early warning and predictive analysis
- Role-based access and daily safety reports

Future Scope

- Advanced AI models and edge AI deployment
- Multi-post control center and mobile app
- Smart-city integration and cloud deployment
- Digital map of all posts and incidents

"SmartSurround transforms environmental monitoring from simple data collection into intelligent sensing, detection, prediction, and safety management."

Thank You

Questions & Discussion